Topology Optimization
Lecture 9

Wave Equations

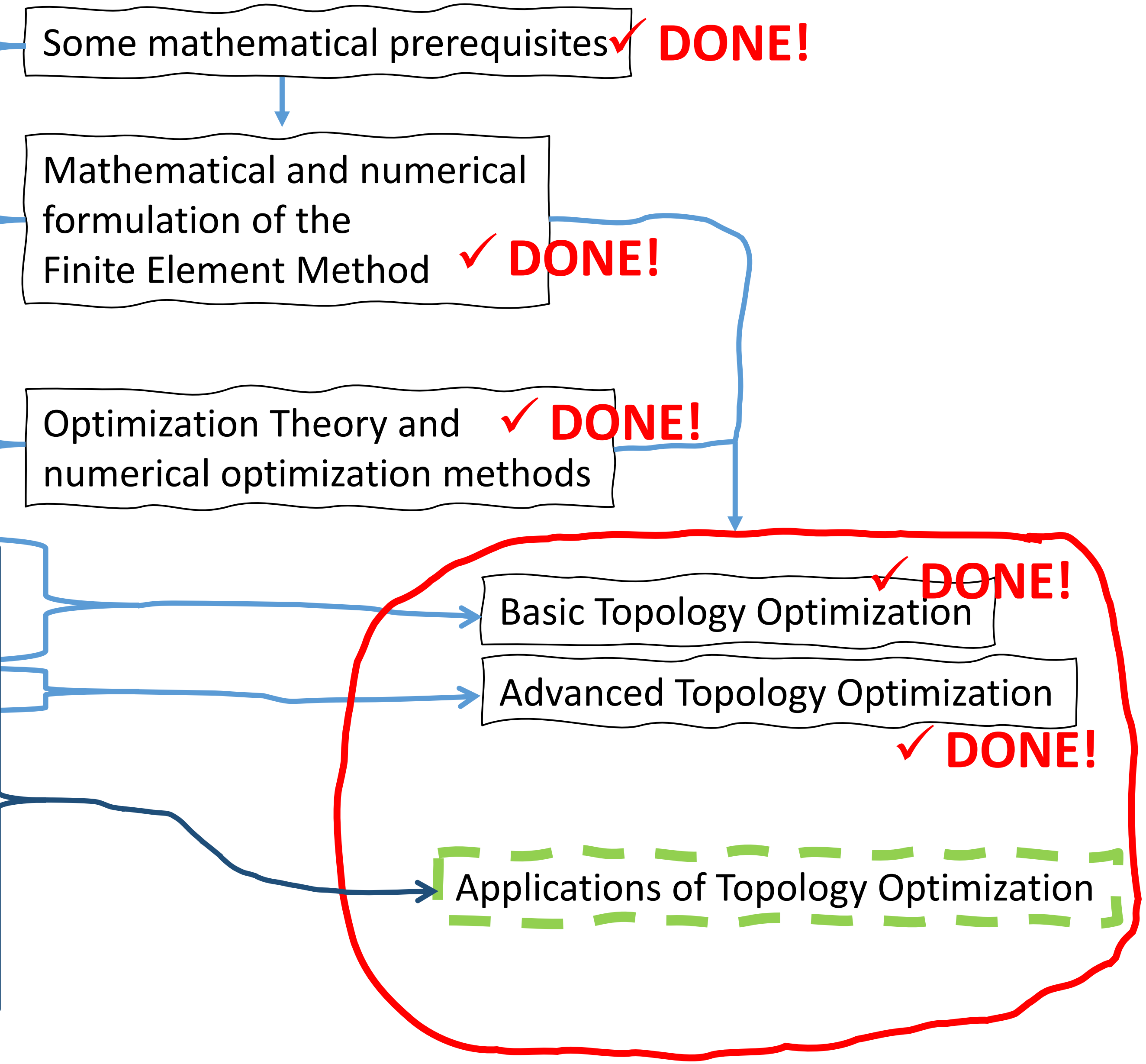
Some references if you want to dig further into this:

1. Fundamentals of General Linear Acoustics, Finn Jacobsen and Peter Moller Juhl, ISBN: 978-1-118-63616-9.

2. Modern Electromagnetism, Andrew Zangwill, ISBN: 978-0-521-89697-9.

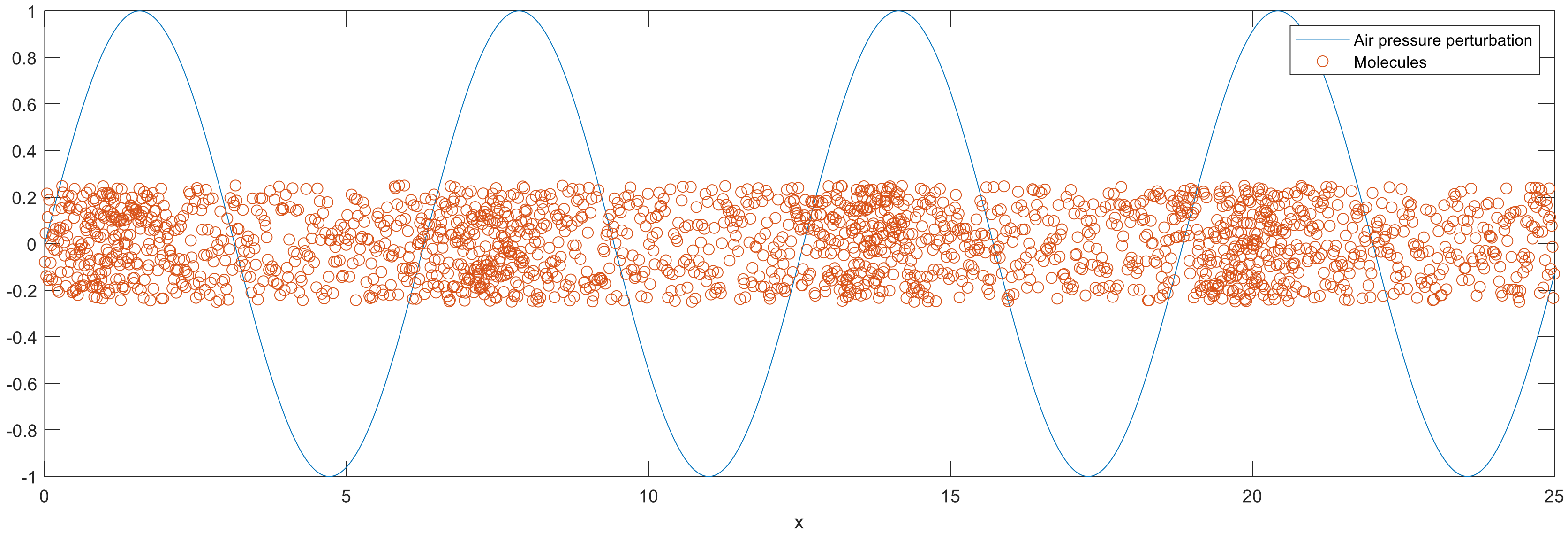
Course overview

No.	Topics
1	Introduction Function and vector spaces, Interpolation, L^2 projection
2	The Finite Element Method for PDEs 1: Poisson's Equation, The Finite Element Method, Properties of the Finite Element Method
3	The Finite Element Method for PDEs 2: Index notation, Linear Elasticity, Non-linear Poisson's equation, Non-linear Elasticity
4	Constrained Optimization 1: Optimality, KKT Theorem, Numerical Optimization Algorithms
5	Constrained Optimization 2: Method of Moving Asymptotes, MPI and PETSc
6	Introduction to Topology Optimization: The idea, Filtering and Projection, Python implementation
7	Gradients
8	Robust topology optimization, More on MPI
9	Wave equations
10	Acoustic waves and more on Robust Optimization
11	Penalty and Augmented Lagrangian Methods Stress-constraints in Topology Optimization 1
12	Stress-constraints in Topology Optimization 2
13	Light concentration using nanoparticle design
14	Heat conduction



Sound Waves

- Sound waves are longitudinal pressure waves travelling in a medium.
- The origin is in the dynamic description of how the medium flows and transmits pressure.
- Sound waves are often considered in gasses (e.g., air) or liquids (e.g., water).
- For solids, both pressure waves (longitudinal) and shear waves (transverse) exist.
- The waves are described, in the linear approximation, by a linear scalar wave equation.
- The solution of the wave equation is the pressure perturbation in space and time due to the sound wave.
- The frequencies of sound waves are typically from 20 Hz to 20 kHz.



Sound Waves

- The fundamental equations of fluid dynamics can be written as

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0$$

$$\frac{\partial \rho \boldsymbol{u}}{\partial t} + \nabla \cdot (\rho \boldsymbol{u} \otimes \boldsymbol{u} + \tau) = -\nabla p + \rho \boldsymbol{g}$$

$$\rho T \frac{\partial s}{\partial t} + \rho T \boldsymbol{u} \cdot \nabla s = \tau : \nabla \boldsymbol{u} - \nabla \cdot \boldsymbol{q}$$

$\boldsymbol{u} \otimes \boldsymbol{u}$ is the tensor product of $\boldsymbol{u}$ with itself.
In index notation $(\boldsymbol{u} \otimes \boldsymbol{u})_{ij} = u_i u_j$.

$\tau : \nabla \boldsymbol{u}$ is the contraction of τ and $\nabla \boldsymbol{u}$.
In index notation $\tau : \nabla \boldsymbol{u} = \tau_{ij} (\nabla \boldsymbol{u})_{ij} = \tau_{ij} u_{i,j}$

  where ρ is the density, $\boldsymbol{u}$ is the velocity, s is the entropy, T is the temperature, $\boldsymbol{g}$ is the gravitational acceleration vector, $\boldsymbol{q}$ is a heat flux, and τ is the deviatoric stress tensor

$$\tau = (\zeta \nabla \cdot \boldsymbol{u}) I + \mu \left(\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^T - \frac{2}{3} (\nabla \cdot \boldsymbol{u}) I \right)$$

  $\zeta = \left(\lambda + \frac{2}{3} \mu \right)$ is the bulk viscosity, λ is the 2nd viscosity and μ is the dynamic viscosity.

- The three equations expresses conservation of mass, momentum and energy, respectively.
- To close the equations, constitutive equations are needed, e.g., $p(\rho, s)$ and $T(\rho, s)$. These are often named the equations of state.

Sound Waves

- The fundamental equations of fluid dynamics can be written as

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0$$
$$\frac{\partial \rho \boldsymbol{u}}{\partial t} + \nabla \cdot (\rho \boldsymbol{u} \otimes \boldsymbol{u} + \tau) = -\nabla p + \rho \boldsymbol{g}$$
$$\rho T \frac{\partial s}{\partial t} + \rho T \boldsymbol{u} \cdot \nabla s = \tau : \nabla \boldsymbol{u} - \nabla \cdot \boldsymbol{q}$$

- Considering small perturbations to a fluid at rest, without gravity and heat flux, we use ‘ for small quantities and write

$$\rho = \rho_0 + \rho', \qquad \boldsymbol{u} = \boldsymbol{u}_0 + \boldsymbol{u}', \qquad p = p_0 + p', \qquad T = T_0 + T', \qquad s = s_0 + s'$$

- To make things simple, we take $\boldsymbol{u}_0 = \boldsymbol{g} = \boldsymbol{q} = 0$ and assume the 0-quantities are constant, viscous effects are unimportant, and we neglect higher orders of the ‘-quantities. The equations then reduce to

$$\frac{\partial(\rho_0 + \rho')}{\partial t} + \nabla \cdot \big((\rho_0 + \rho')\boldsymbol{u}'\big) = 0 \;\Rightarrow\; \boxed{\frac{\partial \rho'}{\partial t} + \nabla \cdot (\rho_0 \boldsymbol{u}') = 0} \text{ as } \frac{\partial \rho_0}{\partial t} + \nabla \cdot (\rho_0 \boldsymbol{u}_0) = \frac{\partial \rho_0}{\partial t} = 0$$

$$\frac{\partial(\rho_0 + \rho')\boldsymbol{u}'}{\partial t} + \nabla \cdot \big((\rho_0 + \rho')\boldsymbol{u}' \otimes \boldsymbol{u}' + \tau(\boldsymbol{u}')\big) = -\nabla(p_0 + p') \;\Rightarrow\; \boxed{\rho_0 \frac{\partial \boldsymbol{u}'}{\partial t} + \nabla p' = 0}$$

$$(\rho_0 + \rho')(T_0 + T')\frac{\partial(s_0 + s')}{\partial t} + (\rho_0 + \rho')(T_0 + T')\boldsymbol{u}' \cdot \nabla(s_0 + s') = 0 \;\Rightarrow\; \rho_0 T_0 \frac{\partial s'}{\partial t} = 0 \;\Rightarrow\; \boxed{\frac{\partial s'}{\partial t} = 0}$$

Sound Waves

$$\frac{\partial \rho'}{\partial t} + \nabla \cdot (\rho_0 \boldsymbol{u}') = 0$$

$$\rho_0 \frac{\partial \boldsymbol{u}'}{\partial t} + \nabla p' = 0$$

$$\frac{\partial s'}{\partial t} = 0$$

- To close the system, an equation of state expressing the pressure is needed:

$$p(\rho, s) \;\; \Rightarrow \;\; dp = \left(\frac{\partial p}{\partial \rho}\right)\bigg|_s d\rho + \left(\frac{\partial p}{\partial s}\right)\bigg|_\rho ds = c^2 d\rho + \left(\frac{\partial p}{\partial s}\right)\bigg|_\rho ds$$

  - $\left(\frac{\partial p}{\partial \rho}\right)\big|_s$ means s is a constant when taking the derivative.
  - $c(\rho, s)$ is the isentropic speed of sound with $c^2 \equiv \left(\frac{\partial p}{\partial \rho}\right)\big|_s$.

- We now write a linear expansion of the pressure near (p_0, s_0)

$$p(\rho, s) \approx p_0 + dp = p_0 + c^2(\rho_0, s_0) d\rho + \left(\frac{\partial p}{\partial s}\right)\bigg|_{\rho=\rho_0, s=s_0} ds$$

- $\frac{\partial s'}{\partial t} = 0$ states that the entropy perturbation is constant, so we may choose $\mathrm{s}' = ds = 0$ to write

$$p(\rho, s) \approx p_0 + c^2(\rho_0, s_0)\rho' = p_0 + p' \;\; \Rightarrow \;\; \boxed{p' = c^2(\rho_0, s_0)\rho' = c^2 \rho'}$$

  where it is understood that the speed of sound is evaluated at the background pressure and entropy.

- This expressed the pressure perturbation, p', due to a density perturbation, ρ'.

Sound Waves

$$\frac{\partial \rho'}{\partial t} + \nabla \cdot (\rho_0 \boldsymbol{u}') = 0$$
$$\rho_0 \frac{\partial \boldsymbol{u}'}{\partial t} + \nabla p' = 0$$
$$\frac{\partial s'}{\partial t} = 0$$

- With s'=0, the equations of motion are

$$\frac{\partial \rho'}{\partial t} + \nabla \cdot (\rho_0 \boldsymbol{u}') = 0 \qquad (*)$$
$$\rho_0 \frac{\partial \boldsymbol{u}'}{\partial t} + \nabla p' = 0 \qquad (**)$$

- Taking the time-derivative of the $(*)$ equation and using $(**)$ and $\rho' = \frac{1}{c^2} p'$ yield

$$\frac{\partial^2 \rho'}{\partial t^2} + \nabla \cdot \left(\rho_0 \frac{\partial \boldsymbol{u}'}{\partial t} \right) = \frac{\partial^2 \rho'}{\partial t^2} - \nabla \cdot \nabla p' = 0 \;\Rightarrow\; \frac{1}{c^2} \frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = 0 \;\Rightarrow\; \frac{\partial^2 p'}{\partial t^2} - c^2 \nabla^2 p' = 0$$

- To ease the notation, the primes are now dropped and the main equations are

$$\frac{\partial^2 p}{\partial t^2} - c^2 \nabla^2 p = 0$$
$$\rho_0 \frac{\partial \boldsymbol{u}}{\partial t} + \nabla p = 0$$

- The first equation is the scalar wave equation for the pressure perturbation and the second equation give the flow velocity perturbation.
- For topology optimization, we need to consider inhomogeneous media. A similar derivation yield a slightly different pressure equation:

$$\frac{1}{\rho_0 c^2} \frac{\partial^2 p}{\partial t^2} - \nabla \cdot \left(\frac{1}{\rho_0} \nabla p \right) = 0 \;\text{ or }\; \frac{1}{K} \frac{\partial^2 p}{\partial t^2} - \nabla \cdot \left(\frac{1}{\rho_0} \nabla p \right) = 0$$

- The quantity $\rho_0 c^2 = K$, where K is the bulk modulus at constant entropy, named the adiabatic bulk modulus. This is valid for gasses and liquids.
- This form of the acoustic wave equation is used when the “background density”, ρ_0, change with position, e.g., when multiple materials are present.
- The speed of sound of pressure waves in solids is found from $K = \rho_0 c^2 = \frac{E(1-\nu)}{(1+\nu)(1-2\nu)}$. K and the speed of sound is thus related to Young’s modulus and Poisson’s ratio.

Sound Waves

- The speed of sound and density enter directly in the wave equation and the velocity equation:

$$\frac{\partial^2 p}{\partial t^2} - c^2 \nabla^2 p = 0$$
$$\rho_0 \frac{\partial \boldsymbol{u}}{\partial t} + \nabla p = 0$$

- The speed of sound varies greatly from gas, to liquid, to solid:
  - In air is ~340 m/s=1224 km/h.
  - In water ~1500 m/s=5400 km/h.
  - In aluminum ~6300 m/s=22680 km/h.
- People also sometimes introduce the following quantities in wave problems:
  - $n = \frac{c_0}{c}$, the refractive index as the ratio between a reference speed c_0 and the wave speed in the medium, c.
  - $Z = c\rho_0$, the impedance of the medium.

Traveling Plane Wave Solution

- A family of solutions to $\frac{\partial^2 p}{\partial t^2} - c^2 \nabla^2 p = 0$ can be written as

$$p = p(\omega t - \boldsymbol{k} \cdot \boldsymbol{r} + \phi) = p\big(\omega t - k_x x - k_y y - k_z z + \phi\big) = p(\xi), \qquad \xi = \omega t - \boldsymbol{k} \cdot \boldsymbol{r} + \phi, \qquad \boldsymbol{k} = \big(k_x, k_y, k_z\big)^T, \qquad c = \frac{\omega}{k}$$

- $\boldsymbol{k}$ is named the wave vector and the size (non-bold) $k = |\boldsymbol{k}|$ is named the wave number.
- ϕ is a constant and ω is the angular frequency.
- To show that it is a solution, one can simply insert into the wave equation:
  - Using $\frac{\partial p}{\partial x}$=$\frac{\partial p}{\partial \xi}\frac{\partial \xi}{\partial x} = p' \frac{\partial \xi}{\partial x} = -k_x p'(\xi)$ etc. with $p' = \frac{\partial p}{\partial \xi} = p'(\xi)$, we get ($p'$ is now the derivative w.r.t. ξ, *not* the perturbation)

$$\nabla p = \begin{bmatrix} \frac{\partial p}{\partial x} \\ \frac{\partial p}{\partial y} \\ \frac{\partial p}{\partial z} \end{bmatrix} = \begin{bmatrix} -k_x p' \\ -k_y p' \\ -k_z p' \end{bmatrix} = -\boldsymbol{k} p'$$

$$\nabla^2 p = \nabla \cdot \nabla p' = -k_x \frac{\partial p'}{\partial x} - k_y \frac{\partial p'}{\partial y} - k_z \frac{\partial p'}{\partial z} = k_x^2 p'' + k_y^2 p'' + k_z^2 p'' = k^2 p'' \text{ with } k^2 = |\boldsymbol{k}|^2$$

  - Inserting into $\frac{\partial^2 p}{\partial t^2} - c^2 \nabla^2 p = 0$ yield:

$$\omega^2 p'' - c^2 k^2 p'' = 0 \ \Rightarrow \ \omega^2 - c^2 k^2 = 0 \ \Rightarrow \ c = \frac{\omega}{k}$$

- $c = \omega / k$ is named the dispersion relation, showing the relation between the wave number, k, the (angular) frequency, ω, and the wave velocity, c.

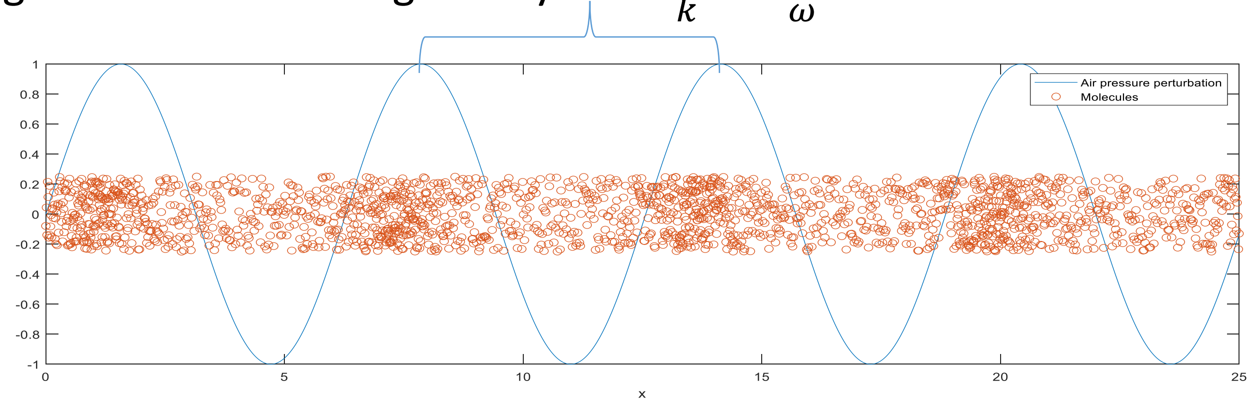
Traveling Plane Wave Solution

- A solution on the form $p = p(\omega t - \boldsymbol{k} \cdot \boldsymbol{r} + \phi) = p(\xi)$ is named a plane wave solution.
- At time t_1, the phase $\xi = \omega t - \boldsymbol{k} \cdot \boldsymbol{r} + \phi$ is constant on a plane with normal vector $\boldsymbol{k}$.
- At a later time, t_2, the plane of constant phase has moved a distance $c(t_2 - t_1) = c\Delta t$ in the direction of the wave vector $\boldsymbol{k}$.
- The wave is traveling with velocity $c = \frac{\omega}{k}$ in the direction of the wave vector $\boldsymbol{k}$.
- Often $p(\xi) \propto \cos(\xi)$ or $p(\xi) \propto \sin(\xi)$ or $p(\xi) \propto e^{i\xi}$ are used in plane wave analysis.
- Note that if $\boldsymbol{k} = (k - i\alpha)\hat{k}$, with $\hat{k}$ being a unit vector, then

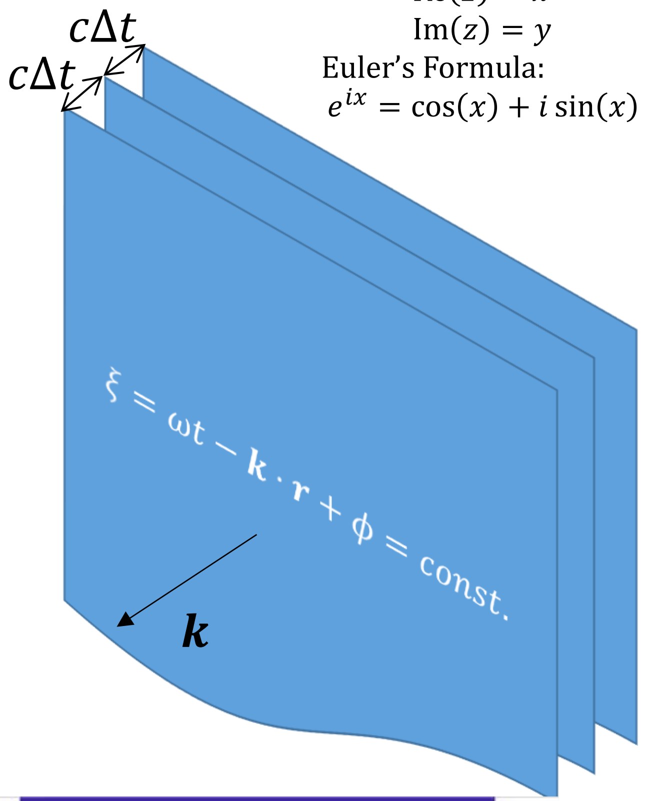
$$p(\xi) \propto e^{i\xi} = e^{i(\omega t - k\hat{k}\cdot\boldsymbol{r} + \phi)} e^{-\alpha\hat{k}\cdot\boldsymbol{r}}$$

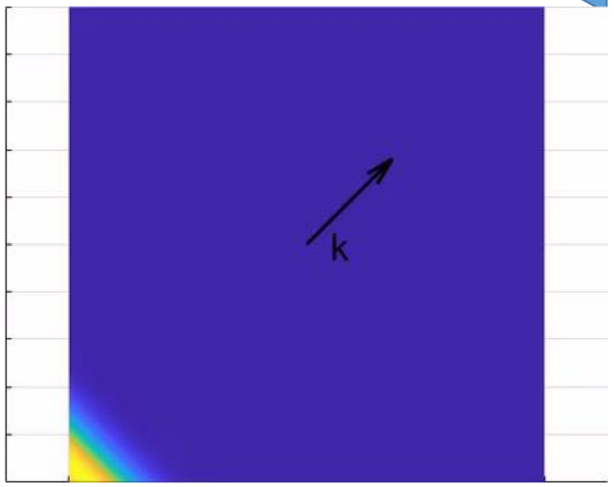
  is decaying in the direction of travel. This can be used to introduce losses in a medium. If you use the opposite sign for the imaginary part, you introduce gain in the medium, which is usually a mistake…

- Using $\xi = \omega t - \boldsymbol{k} \cdot \boldsymbol{r} + \phi$ and the fact that, e.g., $\sin(\xi)$ is periodic with a period of 2π, the wavelength λ is found to be given by $\lambda = \frac{2\pi}{k} = \frac{2\pi c}{\omega}$.



Imaginary numbers:
$i^2 = -1$
$z = x + iy$
$\mathrm{Re}(z) = x$
$\mathrm{Im}(z) = y$
Euler's Formula:
$e^{ix} = \cos(x) + i\sin(x)$





Time-Harmonic Sound Waves

Imaginary numbers:
$i^2 = -1$
$z = x + iy$
$\mathrm{Re}(z) = x$
$\mathrm{Im}(z) = y$
Euler's Formula:
$e^{ix} = \cos(x) + i\sin(x)$

- The pressure and velocity equations are:

$$\frac{\partial^2 p}{\partial t^2} - c^2 \nabla^2 p = 0$$
$$\rho_0 \frac{\partial \boldsymbol{u}}{\partial t} + \nabla p = 0$$

- Sometimes it is sufficient to work with only one frequency, ω.

- In this case, it is convenient to use the complex notation:

$$e^{i\omega t} = \cos(\omega t) + i\sin(\omega t)$$
$$p(\boldsymbol{r}, t) = A\cos(\omega t) = \mathrm{Re}\left(p(\boldsymbol{r}) e^{i\omega t}\right), \qquad \boldsymbol{u}(\boldsymbol{r}, t) = B\cos(\omega t) = \mathrm{Re}\left(\boldsymbol{u}(\boldsymbol{r}) e^{i\omega t}\right)$$

- The time-harmonic equations are obtained as

$$i^2\omega^2 p e^{i\omega t} - c^2\nabla^2 p e^{i\omega t} = 0 \;\; \Rightarrow \;\; c^2\nabla^2 p + \omega^2 p = 0$$
$$i\omega\rho_0 \boldsymbol{u} e^{i\omega t} + \nabla p e^{i\omega t} = 0 \qquad \Rightarrow \qquad i\omega\rho_0\boldsymbol{u} + \nabla p = 0$$

  where $p = p(\boldsymbol{r})$ and $\boldsymbol{u} = \boldsymbol{u}(\boldsymbol{r})$ and both are complex functions.
  The Re part was skipped and can be taken at the very end of the calculation, if needed.

- These equations are static, and therefore much faster to solve than the dynamic equations at the top.

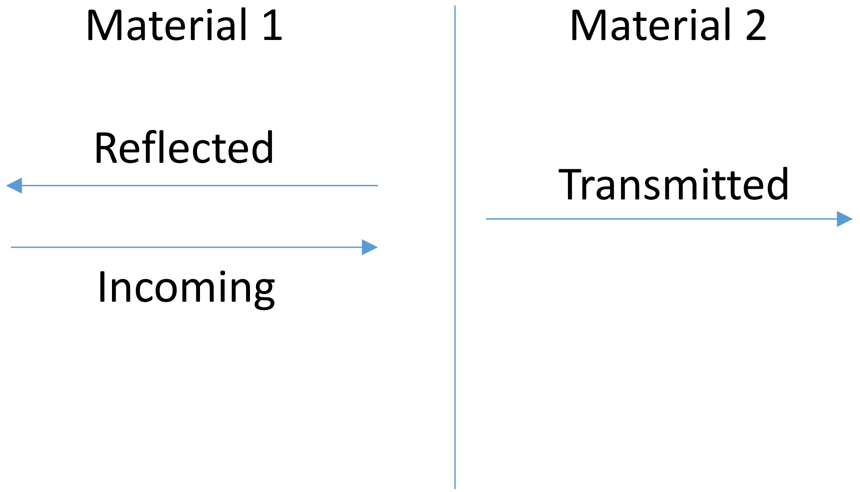
- In inhomogeneous media $\frac{1}{K}\frac{\partial^2 p}{\partial t^2} - \nabla \cdot \left(\frac{1}{\rho_0}\nabla p\right) = 0$, and time-harmonic pressure equation is:

$$\nabla \cdot \left(\frac{1}{\rho_0}\nabla p\right) + \frac{\omega^2}{K} p = 0$$

  This equation is considered in the next paper we will study.

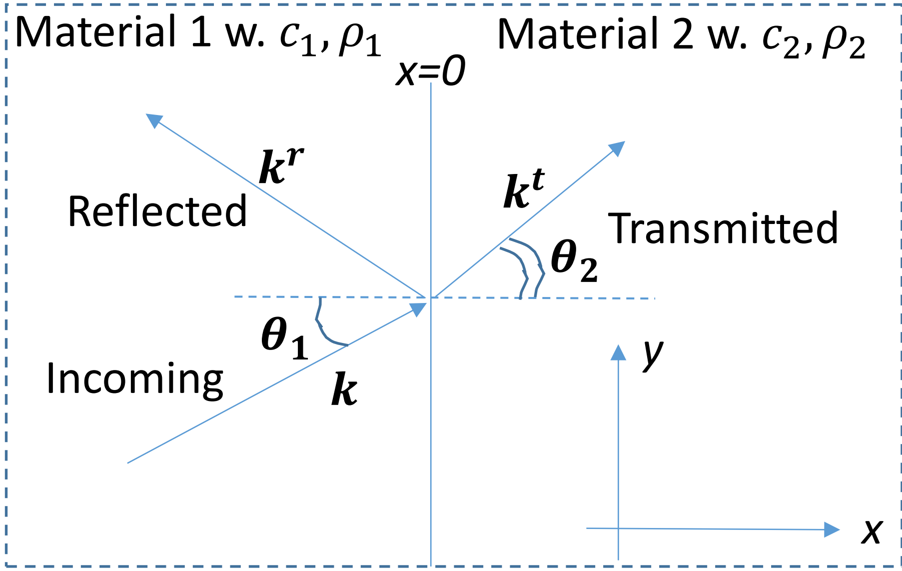
Reflection and Refraction at Interfaces

- When a wave hits an interface between two materials, some part of the wave is reflected, and another part is transmitted across the interface.



- We will now derive some expressions for this process.
- We will use the simplest plane harmonic wave, namely waves on the form $p = Ae^{i(\omega t - \boldsymbol{k} \cdot \boldsymbol{r})}$.

Reflection and Refraction at Interfaces

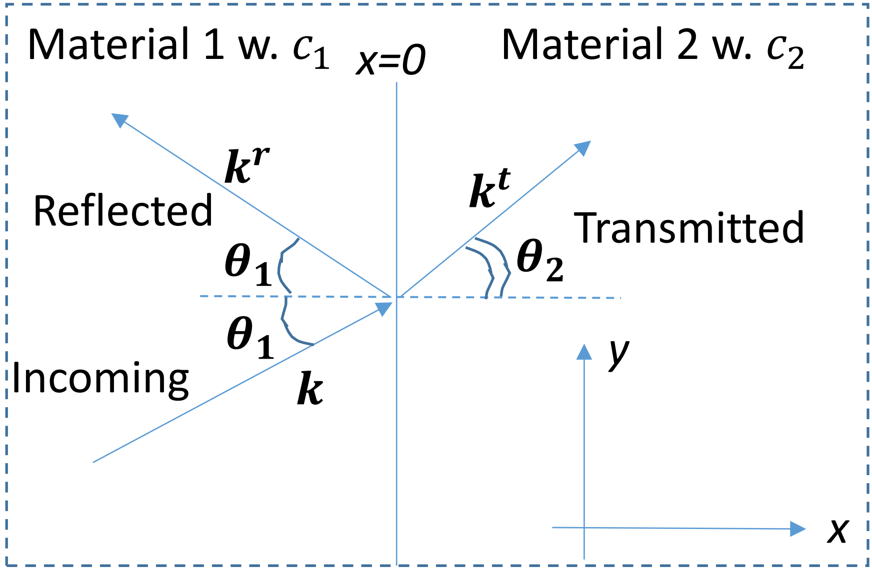


- We take the incoming wave to have an angle θ_1 with the interface normal.
- The interface is at *x*=0.
- For x<0 we write $p = Ie^{i(\omega t - \boldsymbol{k}\cdot\boldsymbol{r})} + Re^{i(\omega t - \boldsymbol{k}^r\cdot\boldsymbol{r})}$ as there is an incoming wave with amplitude I and a reflected wave with amplitude R.
- For x>0 we write $p = Te^{i(\omega t - \boldsymbol{k}^t\cdot\boldsymbol{r})}$ as there is a transmitted wave with amplitude T.
- We write $\boldsymbol{k} = \begin{pmatrix} k_1 \cos\theta_1 \\ k_1 \sin\theta_1 \end{pmatrix}$, $\boldsymbol{k^r} = \begin{pmatrix} k_x^r \\ k_y^r \end{pmatrix}$, $\boldsymbol{k^t} = \begin{pmatrix} k_x^t \\ k_y^t \end{pmatrix}$.
- Demanding the pressure to be equal at the interface, we get
$$Ie^{i(\omega t - \boldsymbol{k}\cdot\boldsymbol{r})}\Big|_{x=0} + Re^{i(\omega t - \boldsymbol{k}^r\cdot\boldsymbol{r})}\Big|_{x=0} = Te^{i(\omega t - \boldsymbol{k}^t\cdot\boldsymbol{r})}\Big|_{x=0} \quad \Rightarrow \quad Ie^{-ik_1\sin\theta_1 y} + Re^{-ik_y^r y} = Te^{-ik_y^t y} \quad \Rightarrow \quad k_1 \sin\theta_1 = k_y^r = k_y^t$$
since it must be true for all values of y.
- Since $c = \frac{\omega}{k}$ for each medium then $|\boldsymbol{k}| = k_1 = \frac{\omega}{c_1}$ and $|\boldsymbol{k^r}| = \frac{\omega}{c_1} = k_1$ and $|\boldsymbol{k^t}| = \frac{\omega}{c_2} = k_2$.
- For $\boldsymbol{k^r} = \begin{pmatrix} k_x^r \\ k_1 \sin\theta_1 \end{pmatrix}$ we get $k_x^r = -k_1\cos\theta_1$ and thus $\boldsymbol{k^r} = \begin{pmatrix} -k_1 \cos\theta_1 \\ k_1 \sin\theta_1 \end{pmatrix}$ - "the incident angle is equal to the reflected angle".
- Writing $\boldsymbol{k^t} = \begin{pmatrix} k_x^t \\ k_y^t \end{pmatrix} = \begin{pmatrix} k_2 \cos\theta_2 \\ k_2 \sin\theta_2 \end{pmatrix} = \begin{pmatrix} k_2 \cos\theta_2 \\ k_1 \sin\theta_1 \end{pmatrix}$ we get Snell's law:
$$k_2 \sin\theta_2 = k_1 \sin\theta_1$$
- Useful to determine θ_2 from θ_1 and the material properties.

Reflection and Refraction at Interfaces

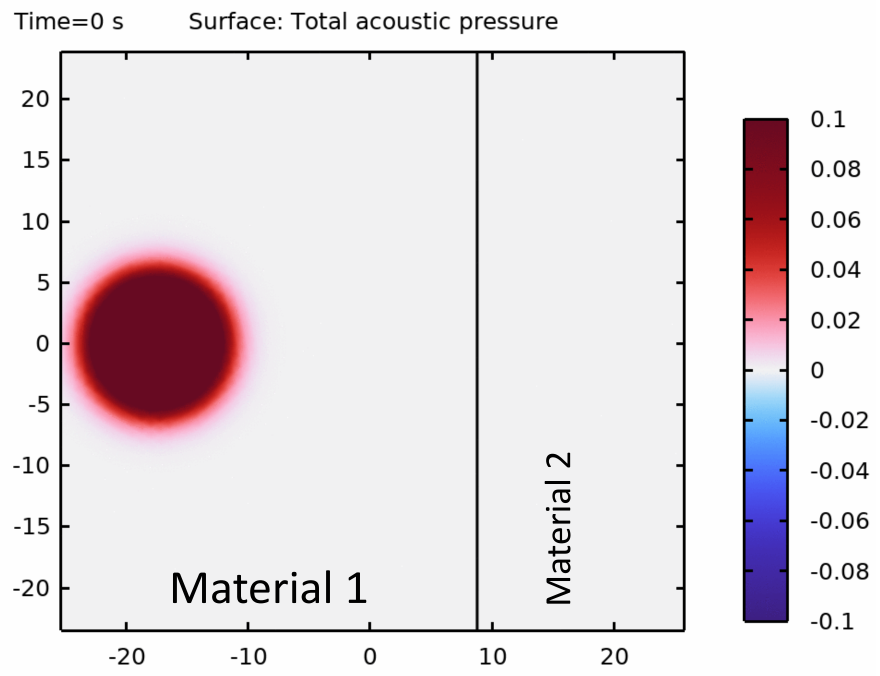
- Snell's law: $k_2 \sin\theta_2 = k_1 \sin\theta_1$ we got from matching the pressure at the material interface.
- The coefficients *R* and *T* can be found by also matching the velocities at the interface using $\rho \frac{\partial \boldsymbol{u}}{\partial t} + \nabla p = 0$.
- The results (derive yourself ☺) can be written as

$$r(\theta_1) = \frac{R}{I} = \frac{Z_2 \cos\theta_1 - Z_1 \cos\theta_2}{Z_2 \cos\theta_1 + Z_1 \cos\theta_2}$$
$$t(\theta_1) = \frac{T}{I} = \frac{2Z_2 \cos\theta_1}{Z_2 \cos\theta_1 + Z_1 \cos\theta_2}$$

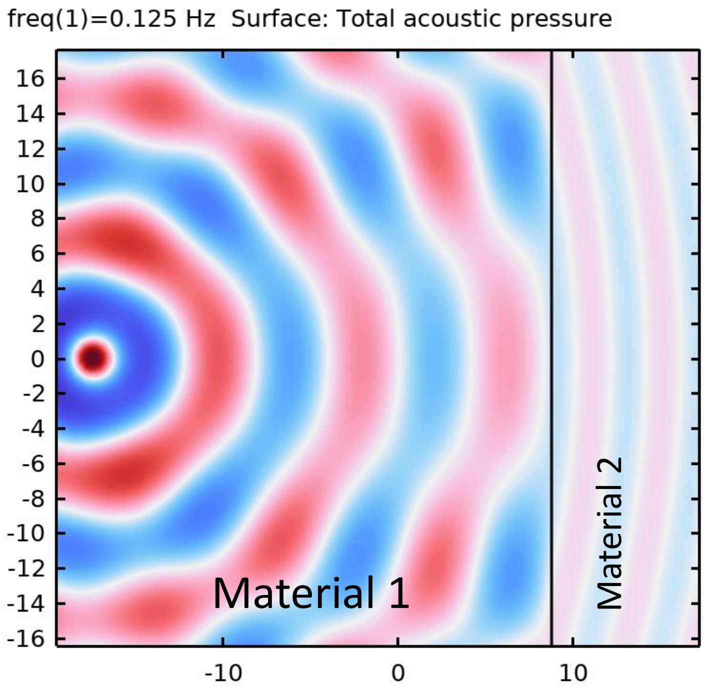


- $Z_1 = \rho_1 c_1$ and $Z_2 = \rho_2 c_2$ are the impedances of the two media.
- In optics, these are known as Fresnel coefficients for TE polarized light.
- For normal incidence, $\theta_1 = 0$ and $r(0) = \frac{Z_2 - Z_1}{Z_2 + Z_1}$.
- If $Z_2 \gg Z_1$ or $Z_1 \gg Z_2$ we get $|r| \approx 1$ and the amplitude of the reflected wave is equal to the amplitude of the incident wave.
- If $Z_2 \approx Z_1$, $|r| \approx 0$ and there is almost no reflected wave. This is named impedance matching.

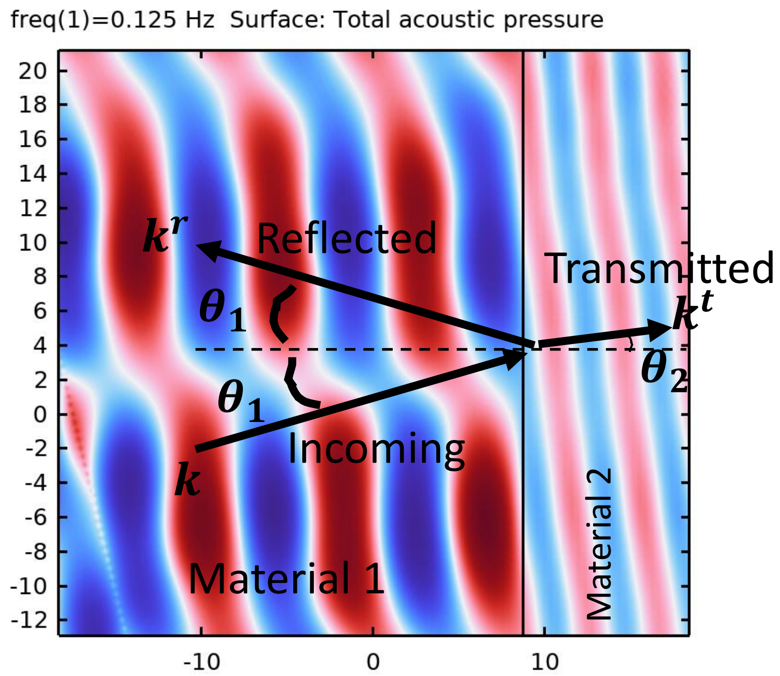
Different views of reflection and refraction at a material interface



Time domain simulation of a pressure pulse.



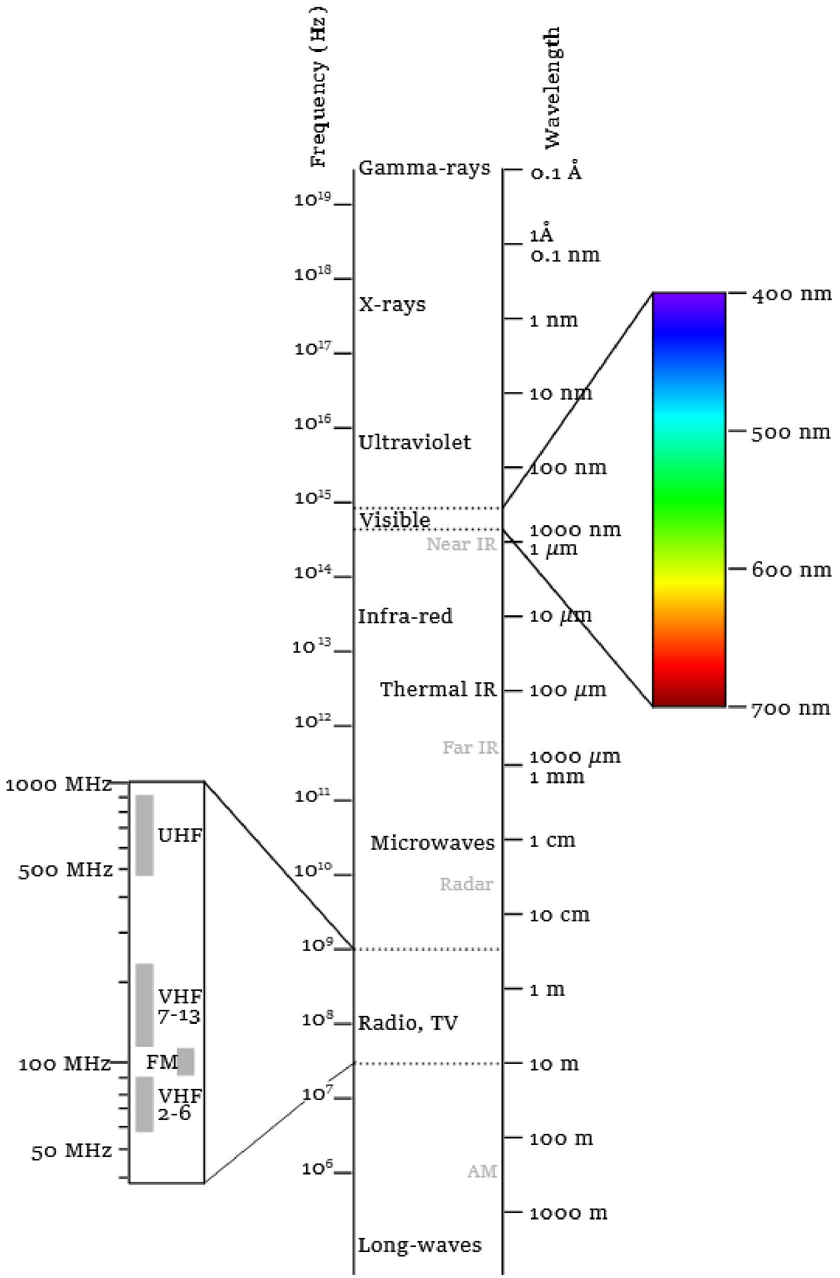
Frequency domain simulation of a time-harmonic pressure point source.



Frequency domain simulation of a time-harmonic plane wave input.

Electromagnetic Waves

- Light waves are waves in electric and magnetic fields.
- Since electric and magnetic field exist both inside and outside materials, the waves also exist in vacuum (e.g., outer space).
- The origin is in Maxwell's equations in vacuum, which work for length scales from below 0.1 Å to the size of the universe.
- Maxwell's equations in matter generally work well "only" from ~1 nm and above due to the constitutive relations.
- In the linear approximation, the waves are described both inside and outside materials by a linear vector wave equation, whose solutions yield the electric field or magnetic field as a function of space and time.
- The frequencies range from below 1 Hz to 10^{25} Hz in the electromagnetic spectrum.



https://en.wikipedia.org/wiki/Electromagnetic_spectrum

Maxwell’s Equations

Maxwell’s equations in matter are

Gauss Law — No Name — Faraday’s law — Amperes law (with Maxwell correction)

$$\nabla \cdot \boldsymbol{D} = \rho_f \quad , \quad \nabla \cdot \boldsymbol{B} = 0 \quad , \quad \nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} \quad , \quad \nabla \times \boldsymbol{H} = \boldsymbol{J}_f + \frac{\partial \boldsymbol{D}}{\partial t}.$$

$\boldsymbol{E}$ is the electric field, $\boldsymbol{D}$ is the electric displacement field, $\boldsymbol{B}$ is the magnetic field and $\boldsymbol{H}$ is the H-field (also named the magnetic field by some...). ρ_f are ”free” charges and $\boldsymbol{J}_f$ are free currents.

To close the equations, constitutive relations are needed. Linear isotropic non-magnetic materials satisfy

$$\boldsymbol{D} = \epsilon \boldsymbol{E} = \epsilon_0 \left(\epsilon' - j\epsilon''\right) \boldsymbol{E} \quad , \quad \mu_0 \boldsymbol{H} = \boldsymbol{B},$$

where $\epsilon = \epsilon_0 \epsilon_r$ and ϵ_r is the complex relative permittivity. It is related to the complex refractive index by

$$N^2 = (n - j\kappa)^2 = \epsilon_r = \frac{c_0^2}{c^2} \quad , \quad c_0 = 1/\sqrt{\mu_0 \epsilon_0} \approx 3 \cdot 10^8 \text{ m/s} \quad , \quad c = 1/\sqrt{\mu_0 \epsilon_r \epsilon_0}$$

The physics in ϵ (and N) is the response of the electrons in the material to an applied field.

If all electrons are tightly bound to the nuclei, the material is a dielectric (e.g., glasses) with $\kappa \approx 0$.

If some electrons can move easily, the material is a metal (e.g., gold) with $\kappa \neq 0$.

The $\kappa \neq 0$ leads to loss of energy in the wave when propagating in the material and high reflections (metals are ”shiny”).

Maxwell's Equations

Maxwell's equations with linear isotropic constitutive relations:

Gauss Law, No Name, Faraday's law, Amperes law (with Maxwell correction)

$$\nabla \cdot \boldsymbol{D} = \rho_f \quad , \quad \nabla \cdot \boldsymbol{B} = 0 \quad , \quad \nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} \quad , \quad \nabla \times \boldsymbol{H} = \boldsymbol{J}_f + \frac{\partial \boldsymbol{D}}{\partial t} \quad ,$$

$$\boldsymbol{D} = \epsilon \boldsymbol{E} \quad , \quad \mu_0 \boldsymbol{H} = \boldsymbol{B}.$$

Taking the curl of Faradays law and using Ampere's law result in

$$\nabla \times (\nabla \times \boldsymbol{E}) = -\frac{\partial \nabla \times \boldsymbol{B}}{\partial t} = -\frac{\partial \nabla \times (\mu_0 \boldsymbol{H})}{\partial t} = -\mu_0 \frac{\partial \boldsymbol{J}_f}{\partial t} - \mu_0 \frac{\partial^2 \boldsymbol{D}}{\partial t^2} \quad \Rightarrow \quad \mu_0 \frac{\partial^2 \epsilon \boldsymbol{E}}{\partial t^2} + \nabla \times (\nabla \times \boldsymbol{E}) = -\mu_0 \frac{\partial \boldsymbol{J}_f}{\partial t}$$

$-\mu_0 \frac{\partial \boldsymbol{J}_f}{\partial t}$ is a source term, showing that if one has a current changing in time, it will generate an electromagnetic wave.

Electric currents are most often due to motion of electrons in materials. Light waves are generated when electrons change energy levels (orbits, shells or bands) in atoms and materials.

Using $\nabla \times (\nabla \times \boldsymbol{A}) = \nabla (\nabla \cdot \boldsymbol{A}) - \nabla^2 \boldsymbol{A}$, one gets

$$\mu_0 \frac{\partial^2 \epsilon \boldsymbol{E}}{\partial t^2} - \nabla^2 \boldsymbol{E} + \nabla (\nabla \cdot \boldsymbol{E}) = -\mu_0 \frac{\partial \boldsymbol{J}_f}{\partial t}$$

The term $\nabla (\nabla \cdot \boldsymbol{E})$ couples the field components, and is not present in the scalar wave equation for sound waves.

Maxwell's Equations

$$\frac{\partial^2 p}{\partial t^2} - c^2 \nabla^2 p = 0$$

The vector wave equation is

$$\mu_0 \frac{\partial^2 \epsilon \boldsymbol{E}}{\partial t^2} - \nabla^2 \boldsymbol{E} + \nabla (\nabla \cdot \boldsymbol{E}) = -\mu_0 \frac{\partial \boldsymbol{J}_f}{\partial t}$$

If $\rho_f = 0$ and ϵ is constant, Gauss law says

$$\nabla \cdot \boldsymbol{D} = \nabla \cdot (\epsilon \boldsymbol{E}) = \epsilon \nabla \cdot \boldsymbol{E} = 0 \;\Rightarrow\; \nabla \cdot \boldsymbol{E} = 0$$

This reduces the vector wave equation to 3 scalar wave equations, one for each component of the electric field

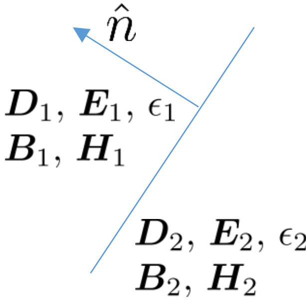
$$\mu_0 \epsilon \frac{\partial^2 \boldsymbol{E}}{\partial t^2} - \nabla^2 \boldsymbol{E} = -\mu_0 \frac{\partial \boldsymbol{J}_f}{\partial t} \;\Rightarrow\; \frac{\partial^2 \boldsymbol{E}}{\partial t^2} - c^2 \nabla^2 \boldsymbol{E} = -\mu_0 c^2 \frac{\partial \boldsymbol{J}_f}{\partial t}$$

where $\mu_0 \epsilon = 1/c^2$ was used. This show that the wave propagates with the velocity c, which is the speed of light in the material.

ϵ is often not constant and one has to solve instead the double-curl vector equation.

Maxwell's Equations

From Maxwell's equations, one can derive so-called interface conditions, for looking at what happens to the fields across a material interface:



$$\hat{n} \cdot (\boldsymbol{D}_1 - \boldsymbol{D}_2) = \sigma_f \quad , \quad \hat{n} \cdot (\boldsymbol{B}_1 - \boldsymbol{B}_2) = 0 \quad , \quad \hat{n} \times (\boldsymbol{E}_1 - \boldsymbol{E}_2) = 0 \quad , \quad \hat{n} \times (\boldsymbol{H}_1 - \boldsymbol{H}_2) = \boldsymbol{K}_f$$

For $\sigma_f = 0$ and $\boldsymbol{K}_f = 0$, the 1st and 3rd equations says: $\hat{n} \cdot (\epsilon_1 \boldsymbol{E}_1) = \hat{n} \cdot (\epsilon_2 \boldsymbol{E}_2)$ and $\hat{n} \times \boldsymbol{E}_1 = \hat{n} \times \boldsymbol{E}_2$. Writing

$$\boldsymbol{E}_1 = \boldsymbol{E}_1^{\perp} \hat{n} + \boldsymbol{E}_1^{||} \hat{m}_1 \quad , \quad \boldsymbol{E}_2 = \boldsymbol{E}_2^{\perp} \hat{n} + \boldsymbol{E}_2^{||} \hat{m}_2$$

with $\hat{m}_i$ tangential to the interface (and perpendicular to $\hat{n}$), it is seen that normal component, $\boldsymbol{E}^{\perp}$, of the electric field is allowed to jump and the tangential component, $\boldsymbol{E}^{||}$, is continuous across a material interface.

These interface conditions leads to the fact, that Lagrange finite element does not work, in general, for electromagnetic wave problems, since they create piecewise continuous functions.

Instead one uses so-called Nédélec elements, which are created for vector fields where only the tangential component (along mesh element boundaries) are continuous.

Nédélec elements are also available in fenics.

They are also named $H(\text{curl})$-conforming finite elements.

Time-Harmonic Electromagnetic Waves

Assuming

$$\boldsymbol{E}(\boldsymbol{r},t) = \boldsymbol{E}(\boldsymbol{r})e^{i\omega t} \quad , \quad \boldsymbol{B}(\boldsymbol{r},t) = \boldsymbol{B}(\boldsymbol{r})e^{i\omega t} \quad , \quad \boldsymbol{J}_f(\boldsymbol{r},t) = \boldsymbol{J}_f(\boldsymbol{r})e^{i\omega t},$$

and inserting into the electric vector wave equation

$$\mu_0 \frac{\partial^2 \epsilon \boldsymbol{E}}{\partial t^2} + \nabla \times (\nabla \times \boldsymbol{E}) = -\mu_0 \frac{\partial \boldsymbol{J}_f}{\partial t}$$

yield the time-harmonic vector wave equation for the electric field:

$$-\nabla \times (\nabla \times \boldsymbol{E}) + \frac{\omega^2}{c^2}\boldsymbol{E} = i\omega\mu_0 \boldsymbol{J}_f.$$

The electric field can oscillate in different directions, which is known as the polarization.

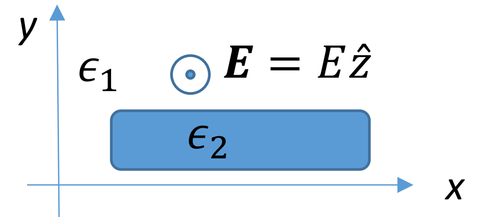
To reduce this to a 2D problem, we assume everything is constant in the z direction and $\boldsymbol{E} = E(x,y)\hat{z}$ and $\boldsymbol{J}_f = J_f\hat{z}$ only have z components. This is named Transverse Electric (TE) polarization. Transverse Magnetic (TM) polarization is also possible.

Then (since $\nabla \cdot \boldsymbol{E} = 0$ and then $\nabla \times (\nabla \times \boldsymbol{E}) = \nabla (\nabla \cdot \boldsymbol{E}) - \nabla^2 \boldsymbol{E} = -\nabla^2 \boldsymbol{E}$)

$$\nabla \times \boldsymbol{E} = \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z}\right)\hat{x} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}\right)\hat{y} + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}\right)\hat{z} = \frac{\partial E_z}{\partial y}\hat{x} - \frac{\partial E_z}{\partial x}\hat{y} \quad \Rightarrow \quad \nabla \times (\nabla \times \boldsymbol{E}) = -\left(\frac{\partial^2 E}{\partial x^2} + \frac{\partial^2 E}{\partial y^2}\right)\hat{z}$$

The time-harmonic TE wave equation thus become

$$\nabla^2 E + \frac{\omega^2}{c^2} E = i\omega\mu_0 J_f$$



Since the electric field is perpendicular to the (x, y) plane, it is transverse to any material boundary in 2D. This $E(x, y)$ is continuous and the equation can be solved with Lagrange finite elements.

The Time-Harmonic Wave Equation

- The TE electric wave equation and the wave equation for sound (pressure) waves can both in the frequency domain be written as (without sources)
$$c^2\nabla^2 f + \omega^2 f = 0$$
- $f = f(x,y)$ is here the z-component of the electric field for TE EM waves.
- $f = p(x,y) = p(\boldsymbol{r})$ in 2D or $f = p(x,y,z) = p(\boldsymbol{r})$ in 3D is the pressure (perturbation) for sound waves.
- f is continuous across material interfaces.
- The time-harmonic solution is $f(\boldsymbol{r})e^{i\omega t}$, with $\boldsymbol{r}$ being the position.
- Many phenomena are thus shared between sound waves and electromagnetic (light) waves and other kinds of waves.